Frontier



Actea Inc

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Actea Inc

Organization location (we welcome applicants from anywhere in the world)

San Francisco, CA

Name(s) of primary point(s) of contact for this application

Jordan Miller, Trond Kristiansen

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Actea Inc is a for-profit Delaware Public Benefits Corporation

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

From Bespoke Models to Scalable MRV for Open-System Marine Carbon Dioxide Removal

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

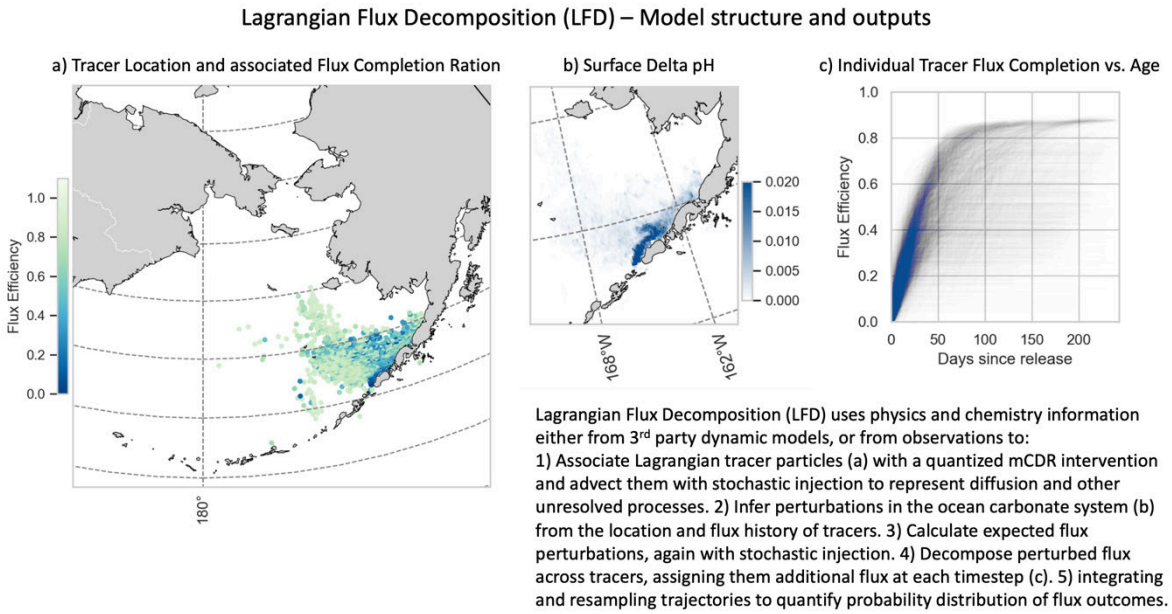
areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Open-system marine Carbon Dioxide Removal (mCDR) has the potential to cost-effectively provide gigaton scale atmospheric CO_2 removal with potential for local ecosystem benefits, such as mitigation of ocean acidification. While the dynamic environment of the ocean enables efficient mCDR, it is also a commons from where key ecosystem services and goods are derived such as fisheries, aquaculture, and recreational sports. Sound management must balance use of the ocean to maintain and improve the health of ecosystems and to continue to deliver broad social and economic value. A step change in mCDR adoption depends on both scalable measurement, reporting, and verification (MRV) and scalable stakeholder trust. For mCDR to achieve widespread adoption, its risks and co-benefits must be communicated clearly and credibly within the environmental analysis context that ocean stakeholders already use for decision-making. This creates a dual challenge: improving the technical foundations of MRV while embedding those insights within real-world ocean monitoring and ecosystem management frameworks.

As identified by Frontier in its description of the priority innovation area of *Improving the MRV toolkit for open-system pathways*, the scalability of current mCDR MRV approaches is limited by cost, complexity, and insufficient understanding of how uncertainty propagates into outcomes. Current approaches depend on computationally expensive ocean models that require significant regional configuration, creating delays, and increasing costs for each new project site. Computational constraints necessitate trade-offs among spatial resolution, geographic coverage, time horizons, the scope of boundary value analysis, and the number of scenarios explored, limiting the ability to fully characterize uncertainty in key outputs such as flux efficiency, and risks or co-benefits from pH change. Data collection efforts may not be cost-optimized due to limited understanding of how measurement uncertainty propagates into MRV outcome uncertainty. As a result, MRV remains expensive, slow to deploy, and difficult to generalize across locations, creating a bottleneck to scaling ocean-based carbon removal and reducing confidence among regulators and stakeholders.

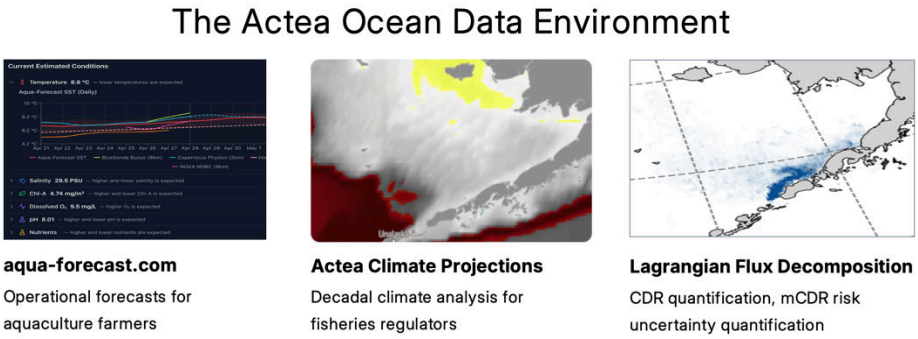
Actea has developed a globally deployable, low-compute MRV framework that reduces computational requirements by approximately 10,000× relative to commonly used ocean models, while enabling uncertainty quantification through ensembles of hundreds of stochastically perturbed re-simulations. This enables robust probability distributions over key outputs such as flux efficiency and pH change. Unlike bespoke dynamic simulations, our approach leverages existing global ocean datasets and statistical methods to efficiently simulate the environmental changes and resulting atmospheric carbon removal from mCDR interventions. The framework also supports a nested multi-resolution architecture, seamlessly coupling

meter-scale near-field models at the intervention site with progressively coarser regional and global ocean models, spanning resolutions from ~1 m to tens of kilometers, to capture both local biogeochemical dynamics and basin-scale tracer dispersal. This allows us to directly quantify uncertainty in carbon removal outcomes given the uncertainty in environmental information, thereby shifting MRV from deterministic outputs to decision-relevant risk assessment.



A key advantage of this approach is its reliance on existing public datasets. Governments and research institutions already produce and maintain operational global ocean observations and model outputs with well-characterized uncertainties. The probabilistic structure of Actea's model can reveal which environmental uncertainties most strongly influence carbon removal outcome uncertainty at any geographic location, enabling targeted data collection where it will most reduce uncertainty. This creates a feedback loop between modeling and measurement, allowing MRV systems to be designed more efficiently while improving confidence in results.

This framework is already built on production ocean analytics infrastructure used by stakeholders today. Actea operates a real-time ocean forecasting platform for aquaculture (https://aqua-forecast.com) and provides temporal and spatial high-resolution ocean climate analytics to fisheries regulators, supporting decisions around water quality, climate impacts, and ecosystem management. Because the same data infrastructure underpins our existing services for fisheries and aquaculture, our MRV framework is already embedded in the environmental context used by ocean stakeholders, enabling clearer communication of risks and co-benefits directly relevant to local communities and users.



Adoption of any mCDR MRV framework as a recognized standard requires that scientific credibility be established through independent validation, quantifiable performance benchmarks, and open scrutiny by the broader research community. This project will rigorously benchmark Actea's MRV model against leading approaches and observational datasets. We will conduct comparisons against the Carbon to Sea MRV database, replicate analyses underlying existing credit issuance frameworks such as those used by Planetary, and collaborate with nonprofit modeling group [C]Worthy to perform high-resolution model intercomparisons. Our initial results (Miller et al. (In review)) show that outputs from currently accepted models, such as flux rate, vary widely between accepted models; our framework's mechanistic interpretability allows for model validation as well as identifying and improving our understanding of the sources of inter-model differences. Results will be published and refined through engagement with the scientific community.

If this work proves Actea's model skillful, it will enable a transition from bespoke, high-cost MRV that can consume 50% of a project budget toward a scalable, model-centric paradigm. Actea will deploy this capability as a software platform, allowing any stakeholder, such as project developers, regulators, or buyers, to commission independent, scientifically grounded MRV analyses on demand, globally and at a fraction of current costs. Pricing could be as low as $1,000 per model year per project, with day-scale turnaround, error bars, and a plan for in-situ observation to narrow uncertainty — orders of magnitude cheaper and faster than today. Because this platform is built within an existing ocean analytics environment, it can simultaneously assess carbon removal performance and ecosystem impacts, including interactions between mCDR, climate-driven ocean acidification, and fisheries-relevant conditions (e.g. habitat, distributions).

By combining scalable MRV with ecosystem analytics, this project addresses both the technical and social barriers to mCDR deployment. It enables not only accurate quantification of carbon removal at accessible cost, but also clearer communication of risks and co-benefits to ocean stakeholders. This integration has the potential to

shift the narrative around mCDR from a perceived trade-off with ocean ecosystems to a tool for protecting and managing them, thereby accelerating both adoption and impact.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

This project aims to rigorously validate and operationalize a novel measurement, reporting, and verification (MRV) framework for open-system mCDR developed by Actea. This framework, the Lagrangian Flux Decomposition (LFD) method (Miller et al. (In review)), provides a reduced-complexity alternative to traditional dynamic ocean models by quantifying intervention-driven changes in air–sea CO_2 flux relative to a counterfactual and attributing those changes to transport and carbonate chemistry processes over time. The model has been implemented in a source-available codebase and is the subject of a pending patent. Early results demonstrate consistency with published models and reveal no fundamental barriers to performance, but do not yet constitute a comprehensive validation with observations. The objective of this project is to establish the level of rigor, transparency, and trust needed for deployment in MRV systems.

This project is structured around four core research questions:

1. Model validity*:* Can the LFD framework reproduce the physical transport dynamics, carbonate system perturbations, and air–sea CO_2 flux estimates produced by high-resolution ocean circulation–biogeochemistry models (e.g., ROMS-based systems)? *Success criterion: integrated 5-year flux efficiency within ±15% of harmonized-forcing ROMS reference, and Bhattacharyya coefficient ≥ 0.75 on plume tracer distributions over 90 days.*

2. Cross model comparison and uncertainty characterization: What drives the differences in outcomes between current MRV models and LFD? How do environmental variability, transport processes, and measurement uncertainty propagate into carbon removal outcomes, and can this be more accurately represented using a probabilistic framework than deterministic approaches? *Success criterion: quantitative attribution of inter-model flux rate discrepancies (currently up to 8×) to specific mechanisms — resolution, parameterization, or boundary conditions — across at least two geographies.*

3. Measurement optimization: Which input variables and their associated uncertainties critically influence MRV outcomes, and how can new information be used to design targeted, efficient environmental monitoring strategies that improve overall confidence in MRV? *Success criterion: published sensitivity ranking of environmental inputs and ≥30% reduction in measurement volume while maintaining equivalent 90% confidence intervals on year-1 flux relative to non-optimized sampling designs.*

4. Ecosystem and stakeholder relevance: Can MRV outputs be integrated into existing ocean monitoring and decision-making systems to contextualize carbon removal within ecosystem states and trends, including ocean acidification and fisheries-relevant conditions? *Success criterion: at least two operational integrations delivered — one with a fisheries regulator and one with an aquaculture stakeholder — providing co-located mCDR and ecosystem analytics.*

Proposed Approach and Methods

Modeling Framework

The LFD framework represents mCDR interventions as ensembles of Lagrangian tracer particles, each carrying a fixed chemical perturbation (Δ Total Alkalinity and/or Δ Dissolved Inorganic Carbon). These tracers are advected using externally supplied ocean circulation and atmospheric forcing fields (e.g., GLORYS ocean reanalysis and ERA5 atmospheric forcing), with stochastic dispersion representing unresolved transport processes. This produces a statistical approximation to advection–diffusion dynamics without requiring re-simulation of ocean physics or biogeochemistry.

At each timestep, particle ensembles are mapped onto an Eulerian grid to reconstruct carbonate system perturbations relative to a counterfactual state. These perturbations are used to compute air–sea CO_2 flux anomalies, which are then redistributed back to particles in a closed-loop accounting framework. This enables direct attribution of atmospheric CO_2 uptake to individual intervention parcels over time, providing a traceable and auditable basis for MRV.

Uncertainty is explicitly represented through stochastic transport sampling, parameterized gas-exchange variability, and sensitivity to environmental inputs. It is quantified using block bootstrap resampling over temporally correlated particle cohorts, ensuring that environmental dependencies are preserved. This structure allows the model not only to estimate carbon removal, but to quantify how uncertainty in environmental inputs translates into uncertainty in outcomes.

Validation and Intercomparison

Initial validation has shown that LFD reproduces key transport and carbonate system dynamics and produces results for carbon removal efficiency and distribution within the range of published models. However, these comparisons have been limited by the availability of compatible datasets and have not yet included controlled, internal-model-state intercomparisons.

This project will conduct a rigorous validation study enabled by new data access and collaborations. In partnership with [C]Worthy, developers of the C-Star modeling system, and leveraging data from ROMS-based simulations used in mCDR research and credit issuance (including Planetary), we will perform direct intercomparisons between LFD and high-resolution Eulerian models. We will additionally compare LFD against SCCWRP's ROMS outputs for LA port waters.M

Validation will be conducted using harmonized or shared environmental forcing fields where possible, allowing models to be evaluated under comparable physical conditions. This enables a more rigorous “apples-to-apples” comparison than is typically possible in MRV validation.

We will evaluate model performance across three levels:

- **Transport dynamics:** comparison of plume evolution, including spatial correlation of surface concentration fields and vertical structure over time. We will run LFD’s Lagrangian tracking using the internal currents from each model in the intercomparison, allowing for calculation of similarity metrics such as the Bhattacharyya coefficient (as calculated in Kristiansen et al. 2026) between LFD under different forcings, and LFD’s plumes in relation to the other models’ plumes.
- **Biogeochemical response:** comparison of carbonate system perturbations (e.g., ΔpH, ΔpCO_2) and resulting air–sea CO_2 flux anomalies
- **Integrated outcomes:** comparison of total carbon uptake and flux efficiency over time

A key differentiator of this approach is the use of typically unexposed model-state comparisons, such as plume structure or current fields, rather than relying solely on final carbon removal estimates. By comparing plume structure and chemical fields directly, we can identify the sources of agreement and divergence between models. Because the LFD framework is computationally efficient, its Lagrangian tracking design provides mechanistic interpretability, and it can be used with forcing fields from different classes of models, it enables systematic diagnosis of how differences in transport representation, resolution, and parameterization influence MRV-relevant outcomes.

While Carbon-to-Sea’s Ocean Alkalinity Enhancement Model Intercomparison Project (OAEMIP) is an attempt to resolve some of these unknowns, it was not

designed to compare regional scale models with global scale models. Our initial work shows that model scale is associated with an 8x difference in simulated mCDR flux rates. To our knowledge, this would be the first intercomparison explicitly designed to evaluate reduced-complexity mCDR models against cross-scale dynamic models using harmonized forcing and intermediate-state diagnostics.

Uncertainty Propagation and Measurement Design

A central component of this project is to link model validation with MRV system design. Using ensemble simulations across a range of environmental conditions and intervention scenarios, we will perform sensitivity analyses to identify which inputs most strongly influence carbon removal outcomes.

This allows us to move beyond simply quantifying uncertainty and instead understand its structure. By identifying dominant uncertainty drivers, we can design targeted data collection strategies that prioritize measurements that can reduce uncertainty in MRV results. This creates synergy between modeling and measurement, enabling more efficient and scalable MRV systems that avoid unnecessary data collection while maintaining high confidence.

Integration with Real-World Ocean Systems

The LFD framework operates directly on global ocean datasets already used in Actea's production environment, including real-time forecasting for aquaculture and climate analytics for fisheries regulators. As part of this project, we will demonstrate how MRV outputs can be integrated with these systems to provide ecosystem-based analysis.

This includes evaluating interactions between multiple mCDR interventions, climate change driven ocean acidification (OA), and fisheries-relevant conditions. For example, we can predict how a proposed mCDR project would change aquaculture farm water chemistry and impact yields. Similarly, projections of multi-decadal interactions between mCDR and climate driven OA contributes to regulatory fisheries planning. Embedding MRV within existing ocean data workflows enables stakeholders to interpret carbon removal in the same context they use to manage environmental risk, improving both usability and trust.

Project Scale and Scope

The study will evaluate multiple geographies and ocean regimes, including regions where high-resolution model data are available and where mCDR deployment is being actively considered. Scenarios will include a range of intervention types (e.g., ocean alkalinity enhancement, Direct Ocean Removal) and environmental conditions, allowing assessment of model performance across diverse and realistic

use cases. Ensemble simulations will be used to capture variability in forcing conditions and intervention parameters, enabling robust statistical characterization of outcomes.

Why This Approach is Best-in-Class

- **No re-simulation of physics or biogeochemistry:**
  Reuses externally generated forcing fields, reducing compute cost by orders of magnitude.
- **Probabilistic, uncertainty-aware framework:**
  LFD explicitly represents uncertainty propagation, informing data collection design.
- **Traceable, parcel-level attribution:**
  Closed-loop particle accounting attributes carbon removal to specific intervention parcels over time, supporting transparency and auditability.
- **Scalable and globally deployable:**
  No regional configuration required, applicable across geographies using existing global datasets.
- **Rigorous validation strategy:**
  This project uses internal-model-state intercomparisons with high-resolution models (ROMS/C-Star), going beyond output-based validation.
- **Integration with stakeholder systems:**
  The framework is interoperable with ocean analytics platforms, contextualizing MRV within ecosystem management workflows.

Outcome

This project aims to show that MRV for open-system mCDR can be both scientifically rigorous and operationally scalable. The result will be a validated framework that anyone can access at relatively low cost, while improving MRV interpretability and trustworthiness for broader mCDR deployment.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

The primary risks in this project relate to data access for validation, model performance under rigorous comparison, ambiguity in model intercomparison, and stakeholder adoption.

1. Data access for model intercomparison: A core component of this project is validation against high-resolution ocean models (e.g., ROMS/C-Star) using harmonized forcing and intermediate-state comparisons. This requires access to

detailed model outputs that are not always publicly available. There is a risk that partner data access is delayed or incomplete.

This risk is mitigated through a diversified collaboration strategy. We are engaging multiple partners, including nonprofit modeling groups (e.g., [C]Worthy) and commercial actors, and have allocated >25% of the project budget to support partner participation and data sharing. We will also incorporate publicly available datasets and published results (e.g., Carbon to Sea MRV database, OAE Atlas outputs, and ROMS-based studies). While any one group's inability to participate may shift the target of comparisons, it should not prevent meaningful validation or publication.

2. Model performance and generalization: There is a risk that the Lagrangian Flux Decomposition (LFD) framework does not sufficiently replicate trusted high-resolution model outputs, or that its performance varies across geographies or conditions in ways that limit its applicability for MRV.

This risk is partially mitigated by initial results, currently under peer review, which show that LFD reproduces key dynamics and falls within the range of existing models. The project explicitly evaluates performance across multiple regions and conditions using ensemble simulations. Even if limitations are identified, the results will provide valuable insight into where reduced-complexity models are reliable, an outcome that is itself a publishable contribution to the open-system MRV literature.

3. Validation ambiguity and model disagreement: Differences between existing high-resolution models may limit the ability to define a clear validation benchmark. Prior work suggests that accepted mCDR models can produce materially different results under similar conditions.

We mitigate this by focusing on intermediate-state comparisons and uncertainty attribution. By comparing plume structure, transport dynamics, and carbonate perturbations, not just final carbon removal, we can diagnose sources of agreement and divergence, framing validation as understanding model behavior rather than as pass/fail comparison. Where disagreement persists, these diagnostics can guide targeted physical sampling to better constrain uncertainty and establish empirical benchmarks for MRV.

4. Stakeholder interpretation and adoption: Even with strong validation, MRV outputs may be difficult for stakeholders to interpret or trust, particularly for regulators and local ocean users.

This risk is mitigated by embedding MRV outputs within Actea's existing ocean analytics systems used by fisheries regulators and aquaculture operators. Presenting results alongside familiar environmental indicators improves interpretability and

supports trust.

5. Execution risk: Execution risk is low. The core modeling framework is already implemented and has produced initial results currently under peer review. The remaining work, including validation, intercomparison, and integration, is within the team's demonstrated capabilities.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	a. MRV model validated against Carbon to Sea, Planetary (ROMS), and C-Worthy (C-Star) reference datasets b. Cross-model comparison dataset enabling controlled evaluation of transport, chemistry, and flux dynamics c. Two peer-reviewed publications documenting model validation performance and inter-model transport and flux discrepancies d. Uncertainty quantification and sensitivity analysis framework for mCDR MRV e. Prototype MRV SaaS platform enabling on-demand global project evaluation
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	These deliverables address a core bottleneck in open-system mCDR: the lack of validated, scalable, and trusted MRV. By providing the first controlled intercomparison across model resolutions and approaches, this work reduces uncertainty in carbon removal estimates and clarifies sources of model disagreement. The resulting framework enables

	orders-of-magnitude faster evaluation of projects, supporting improved uncertainty quantification and more efficient monitoring design. In parallel, the transition from bespoke modeling to a globally deployable MRV platform unlocks scalability, enabling consistent evaluation across geographies and deployment strategies. By embedding MRV within ocean data systems used by regulators and stakeholders, this approach also improves model interpretability and supports the development of social license. Together, these advances reduce cost, increase confidence in net removal, and accelerate the deployment of mCDR toward gigaton-scale potential.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	Comparisons to existing datasets first submitted by month 6. Generation of novel datasets by [C]Worthy in first 6-9 months. Comparisons with [C]Worthy outputs – Completed month 12. Integration with ocean stakeholder platform – by end of month 18.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$450,000